



Networkwalks Academy

CYBERSECURITY • ETHICAL HACKING • NETWORK TRAINING

CYBERSECURITY PROFESSIONAL PROGRAM

Week 02 Report

Reconnaissance & Footprinting Techniques

Applying OSINT, Domain Footprinting, Email/Host Harvesting, and Network Port Scanning to Profile a Target Domain using Kali Linux, Maltego, theHarvester, and Zenmap

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Executive Summary

This report documents Week 02 of the Cybersecurity Professional program at Networkwalks Academy, under the guidance of Instructor Waqas Karim. Building on the isolated Kali Linux lab established in Week 01, this project applied core reconnaissance and footprinting techniques used in the early stages of a penetration test: gathering public information about a target domain before any active exploitation is attempted.

The work covered four modules. First, multiple Kali Linux command-line tools (whois, whatweb, nslookup, curl, wafw00f, and dnsrecon) were used to footprint the domain networkwalks.com, revealing registration details, web technology stack, DNS records, and web application firewall protection. Second, Maltego was installed and used to run OSINT transforms against the target domain and a named individual, graphically mapping related email addresses, social media profiles, and web presence. Third, theHarvester was used to harvest emails and hosts for two targets, illustrating both a successful large-scale result and a limitation encountered when a target returns no data from a given source. Finally, Zenmap (the Nmap GUI) was installed on the local Windows host and used to run Quick Scan and Ping Scan profiles against a local network target, identifying open ports and host availability.

By the end of Week 02, a broad footprint of the target domain had been assembled entirely from open-source and network-scanning techniques, without any exploitation being attempted. This report captures each tool used, the commands run, the results obtained, and what those results mean from a reconnaissance and defensive perspective.

Learning Objectives

The objectives set out for this project were to:

- Understand footprinting and reconnaissance as the first phase of the penetration-testing lifecycle, and why it is performed before any active or intrusive testing.
- Gain hands-on experience with multiple Kali Linux command-line reconnaissance tools, and understand what category of information each one reveals (registration data, web technology, DNS, WAF presence).
- Learn to use Maltego to run OSINT transforms and visually map relationships between a domain, its email addresses, and associated web/social presence.
- Use theHarvester to collect emails and hostnames for a target from public search sources, and understand how results can vary depending on the target and data source selected.
- Install and use Zenmap to run structured Nmap scan profiles (Quick Scan, Ping Scan) and interpret open ports and host-discovery results.
- Understand what discovered information (open SMB/NetBIOS ports, WAF presence, DNS/SPF records, exposed metadata) means from a security and risk perspective.
- Continue building professional technical documentation skills for cybersecurity project work.

1. Introduction

Following the lab environment built in Week 01, this second project moved into the first practical phase of penetration testing: reconnaissance and footprinting. Footprinting is the process of gathering as much information as possible about a target — domain registration, DNS configuration, web technologies, exposed hosts, email addresses, and open network ports — using only passive or low-impact techniques, before any exploitation is attempted. This information forms the basis for planning later, more active stages of a penetration test.

This report documents the complete process across the four modules selected for Week 02: multi-tool footprinting from Kali Linux, OSINT graph reconnaissance with Maltego, email and host harvesting with theHarvester, and network port scanning with Zenmap. For each module, the tool used, the exact commands run, the results obtained, and their security significance are documented.

1.1 Project Scope

	Scope Item	Report Section
✓	Multi-tool footprinting of the target domain using Kali Linux (whois, whatweb, nslookup, curl, wafw00f, dnsrecon)	§2.1
✓	OSINT reconnaissance and relationship mapping using Maltego	§2.2
✓	Email and host harvesting using theHarvester	§2.3
✓	Network port scanning and host discovery using Zenmap	§2.4
—	GHDB-based footprinting attacks (Google Hacking Database)	<i>Not selected</i>
✓	Document challenges faced and lessons learned	§3 – §4

2. Reconnaissance & Footprinting Work

This section documents, tool by tool, the reconnaissance and footprinting work carried out against the target domain networkwalks.com, followed by the network-scanning work carried out with Zenmap against a local network target.

2.1 Multi-Tool Footprinting with Kali Linux (W2-PM1)

Six command-line reconnaissance tools available by default in Kali Linux were used against networkwalks.com, each revealing a different category of information about the target: domain registration, web technology stack, DNS resolution, raw HTTP response data, web application firewall (WAF) presence, and detailed DNS enumeration.

2.1.1 WHOIS Lookup

The whois command was run against networkwalks.com to retrieve public domain registration details.

```
(root@kali) ~/# whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2905
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-08-19T18:10:59Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' (VeriSign) Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
```

Figure 2.1 — whois lookup for networkwalks.com, showing registrar (GoDaddy), creation/expiry dates, and name servers.

The lookup confirmed the domain is registered through GoDaddy, was created in November 2019 with an expiry date in November 2027, and resolves through HostGator name servers (ns6135/ns6136.hostgator.com). All standard ICANN client-transfer/update/delete/renew locks were shown as enabled, indicating the domain is protected against unauthorized transfer.

2.1.2 Web Technology Fingerprinting with WhatWeb

whatweb was used to fingerprint the web technologies running on the target site.

```
(root@kali) ~/# whatweb https://networkwalks.com
https://networkwalks.com [200 OK] Apache, Bootstrap[7.0.4], Cookies[[_wpdm_client]], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google
-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Mana
ger 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IP18&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,s
peculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-ng
inx-cache], WordPress[7.0.4]
```

Figure 2.2 — whatweb output for https://networkwalks.com identifying the server stack and page metadata.

The scan identified an Apache web server, WordPress 7.0.4 (via WPDM Client cookie and the WordPress Download Manager plugin), the site's public IP address (192.232.216.135), jQuery 3.7.1, and Google Tag Manager — giving a clear picture of the underlying content management system and third-party components an attacker could target.

2.1.3 DNS Resolution with nslookup

nslookup was used to resolve the domain against Google's public DNS resolver (8.8.8.8) to confirm the authoritative IP address.

```
(root@kali)-[/home/userhafsa]
# nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135

(root@kali)-[/home/userhafsa]
```

Figure 2.3 — nslookup resolving networkwalks.com to 192.232.216.135 via 8.8.8.8.

2.1.4 HTTP Response Retrieval with curl

curl was used to retrieve the raw HTTP response from the target site.

```
(root@kali)-[/home/userhafsa]
# curl I https://networkwalks.com
curl: (6) Could not resolve host: I
<!DOCTYPE html><html lang="en-US"><head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="profile" href="https://gmpg.org/xfn/11">

  <meta name="robots" content="index, follow, max-image-preview:large, max-video-preview:-1" />

  <!-- This site is optimized with the Yoast SEO plugin v27.9 - https://yoast.com/product/yoast-seo-wordpress/ -->
  <title>Networkwalks Academy</title>
  <meta name="description" content="We specialise in Network & IT training courses including Cisco CCNA, CCNP, Cybersecurity, Ethical Hacking, Python Programming & Linux." />
  <link rel="canonical" href="https://networkwalks.com/" />
  <meta property="og:locale" content="en_US" />
  <meta property="og:type" content="website" />
  <meta property="og:title" content="Networkwalks Academy" />
  <meta property="og:description" content="We specialise in Network & IT training courses including Cisco CCNA, CCNP, Cybersecurity, Ethical Hacking, Python Programming & Linux." />
  <meta property="og:url" content="https://networkwalks.com/" />
  <meta property="og:site_name" content="Networkwalks Academy" />
  <meta property="article:publisher" content="https://www.facebook.com/NetworkWalks/" />
  <meta property="article:modified_time" content="2026-08-04T03:11:27+00:00" />
  <meta property="og:image" content="https://networkwalks.com/wp-content/uploads/2019/12/Network-Walks-Logo-v1.1-small-PNG_200x200.png" />
  <meta property="og:image:width" content="200" />
  <meta property="og:image:height" content="200" />
  <meta property="og:image:type" content="image/png" />
  <meta name="twitter:card" content="summary_large_image" />
  <meta name="twitter:site" content="@networkwalks" />
  <script type="application/ld+json" class="yoast-schema-graph">{"@context":"https://schema.org","@graph":[{"@type":"WebPage","@id":"https://networkwalks.com/","url":"https://networkwalks.com/","name":"Networkwalks Academy","isPartOf":{"@id":"https://networkwalks.com/#website"},"about":{"@id":"https://networkwalks.com/#organization"},"primaryImageOfPage":{"@id":"https://networkwalks.com/#primaryimage"},"image":{"@id":"https://networkwalks.com/#primaryimage"},"thumbnailUrl":"https://networkwalks.com/wp-content/uploads/2019/12/Network-Walks-Logo-v1.1-small-PNG_200x200.png","datePublished":"2019-11-17T13:06:21+00:00","dateModified":"2026-08-04T03:11:27+00:00","description":"We specialise in Network & IT training courses including Cisco CCNA, CCNP, Cybersecurity, Ethical Hacking, Python Programming & Linux.", "breadcrumb":{"@id":"https://networkwalks.com/#breadcrumb"},"inLanguage":"en-US","potentialAction":{"@type":"ReadAction","target":{"@id":"https://networkwalks.com/"},"@type":"ImageObject","inLanguage":"en-US","@id":"https://networkwalks.com/#primaryimage"},"url":"https://networkwalks.com/wp-content/uploads/2019/12/Network-Walks-Logo-v1.1-small-PNG_200x200.png","contentUrl":"https://networkwalks.com/wp-content/uploads/2019/12/Network-Walks-Logo-v1.1-small-PNG_200x200.png","width":200,"height":200,"@type":"BreadcrumbList","@id":"https://networkwalks.com/#breadcrumb","itemListElement":[{"@type":"ListItem","position":1,"name":"Home"}]},{"@type":"WebSite","@id":"https://networkwalks.com/#website","url":"https://networkwalks.com/","name":"Networkwalks Academy","description":"LEARN CERTIFY CONQUER","publisher":{"@id":"h
```

Figure 2.4 — curl retrieving the homepage response from https://networkwalks.com.

The response revealed extensive page metadata: the site is optimized with the Yoast SEO plugin (v27.9), confirms WordPress as the CMS, and exposes Open Graph and Twitter Card metadata along with a JSON-LD schema block describing the site structure — all useful context for an attacker building a profile of the target's technology stack.

2.1.5 Web Application Firewall Detection with WAFW00F

wafw00f was used to check whether the target site is protected by a web application firewall.

```

(root@kali)~[/home/userhafsa]
# wafw00f https://networkwalks.com

WOOF!

404 Hack Not Found
405 Not Allowed
403 Forbidden
502 Bad Gateway
500 Internal Error

~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[*] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[*] Number of requests: 2

(root@kali)~[/home/userhafsa]
#

```

Figure 2.5 — wafw00f identifying ModSecurity (SpiderLabs) protecting <https://networkwalks.com>.

The scan confirmed the site is protected by ModSecurity (SpiderLabs), an open-source WAF. This is a significant finding for a penetration tester, since it indicates that aggressive or malformed requests during later, more active testing phases may be blocked or logged, and that testing traffic should be paced accordingly.

2.1.6 DNS Enumeration with dnsrecon

dnsrecon was used to perform a general DNS enumeration against the domain, going beyond a single-record nslookup query.

```

(root@kali)~[/home/userhafsa]
# dnsrecon -d networkwalks.com
2026-08-19T14:21:29.121178-0400 INFO Starting enumeration for domain: networkwalks.com
2026-08-19T14:21:29.121607-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-08-19T14:21:29.695190-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-08-19T14:21:30.057161-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-08-19T14:21:30.149389-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-08-19T14:21:30.157703-0400 ERROR Recursion enabled on NS Server 192.232.216.131
2026-08-19T14:21:30.168533-0400 INFO Bind Version for 192.232.216.131 "9.11.26-RedHat-9.11.26-6.el8"
2026-08-19T14:21:30.168773-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-08-19T14:21:30.175763-0400 ERROR Recursion enabled on NS Server 50.87.144.87
2026-08-19T14:21:30.184023-0400 INFO Bind Version for 50.87.144.87 "9.11.26-RedHat-9.11.26-6.el8"
2026-08-19T14:21:31.043553-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-08-19T14:21:31.345939-0400 INFO A networkwalks.com 192.232.216.135
2026-08-19T14:21:32.225827-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemniZDNV4q75X-Ij-mjgEeg-UsYI
2026-08-19T14:21:32.226411-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-08-19T14:21:32.809965-0400 INFO Enumerating SRV Records
2026-08-19T14:21:36.899300-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-08-19T14:21:36.899847-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-08-19T14:21:36.900308-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-08-19T14:21:36.900932-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-08-19T14:21:36.901264-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-08-19T14:21:36.901571-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-08-19T14:21:36.901929-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-08-19T14:21:36.902228-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-08-19T14:21:36.903835-0400 INFO 8 Records Found
2026-08-19T14:21:36.904352-0400 INFO Completed enumeration for domain: networkwalks.com

(root@kali)~[/home/userhafsa]
#

```

Figure 2.6 — dnsrecon enumerating SOA, NS, MX, A, TXT, and SRV records for networkwalks.com.

The enumeration returned the domain's SOA and NS records, its mail server (mail.networkwalks.com), an SPF TXT record authorizing its mail infrastructure, a Google site-verification TXT record, and eight Autodiscover SRV records pointing to cPanel's email-discovery service — confirming the domain uses cPanel-based email hosting alongside its WordPress front end.

Result

Six independent Kali Linux tools were used to build a layered footprint of networkwalks.com: registrar and domain-lock data (whois), CMS and web-stack fingerprinting (whatweb), authoritative DNS resolution (nslookup), raw HTTP/metadata exposure (curl), WAF presence (wafw00f), and full DNS record enumeration (dnsrecon) — together giving a comprehensive, passive picture of the target's infrastructure.

2.2 OSINT Reconnaissance with Maltego (W2-PM3)

Maltego Graph was downloaded and installed to perform OSINT-based footprinting attacks, using its transform engine to automatically discover and graph relationships between a target domain, its email addresses, and its wider web/social presence.

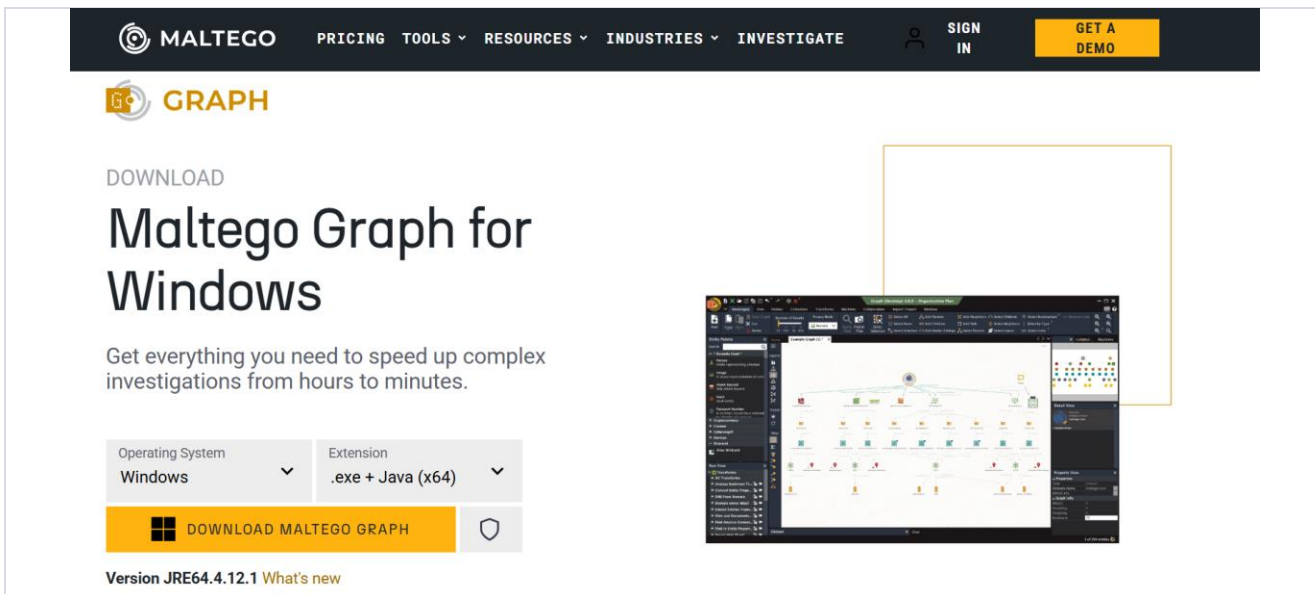


Figure 2.7 — Downloading Maltego Graph for Windows from the official Maltego website.

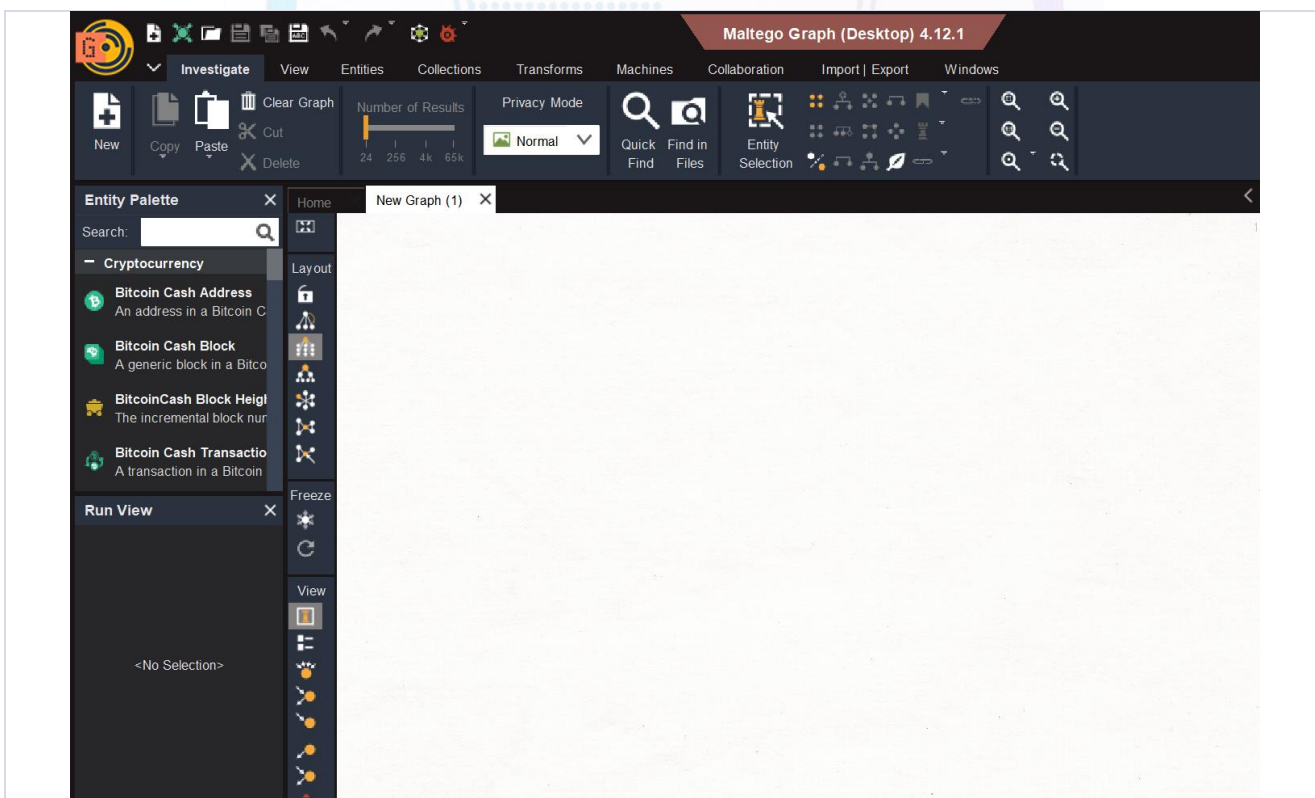


Figure 2.8 — Maltego Graph (Desktop) 4.12.1 launched successfully, showing the Investigate ribbon and empty entity palette.

With Maltego installed and licensed, a new graph was created and a Domain entity for networkwalks.com was seeded. The “To Email Addresses [Search Engine]” transform was run against the domain to discover associated email addresses.

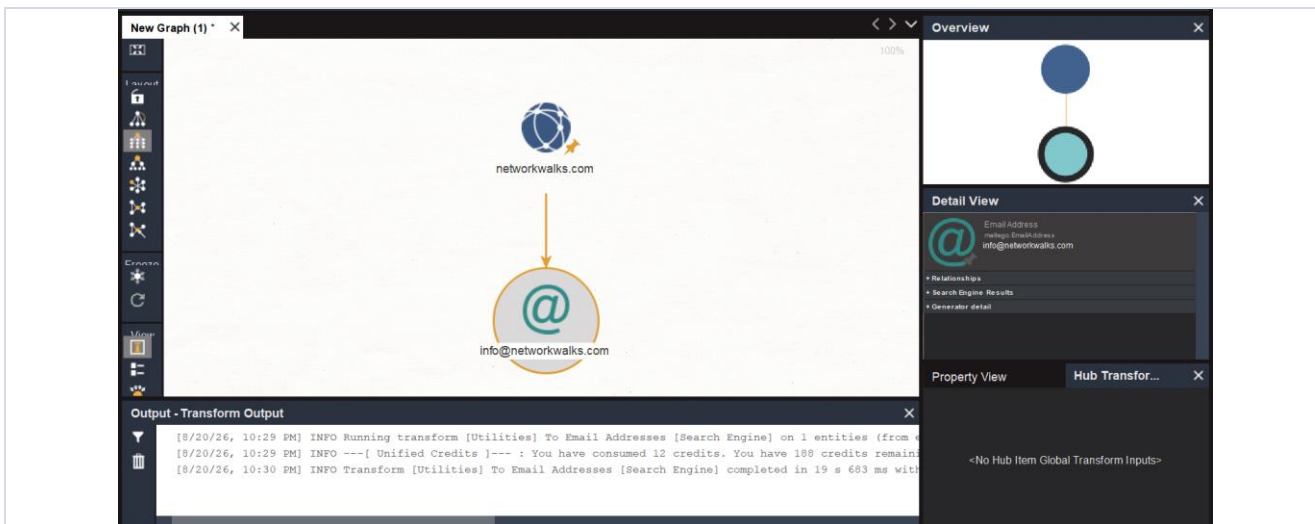


Figure 2.9 — Running the “To Email Addresses” transform on networkwalks.com, discovering info@networkwalks.com.

The transform successfully linked the domain to its published contact address, info@networkwalks.com, confirming the same address seen earlier in the whatweb metadata (§2.1.2). A second graph was then built around a Person entity representing the course instructor, Waqas Karim CCIE, and the “Search Web” and “To Email Addresses [PGP]” transforms were run to map his public web footprint.

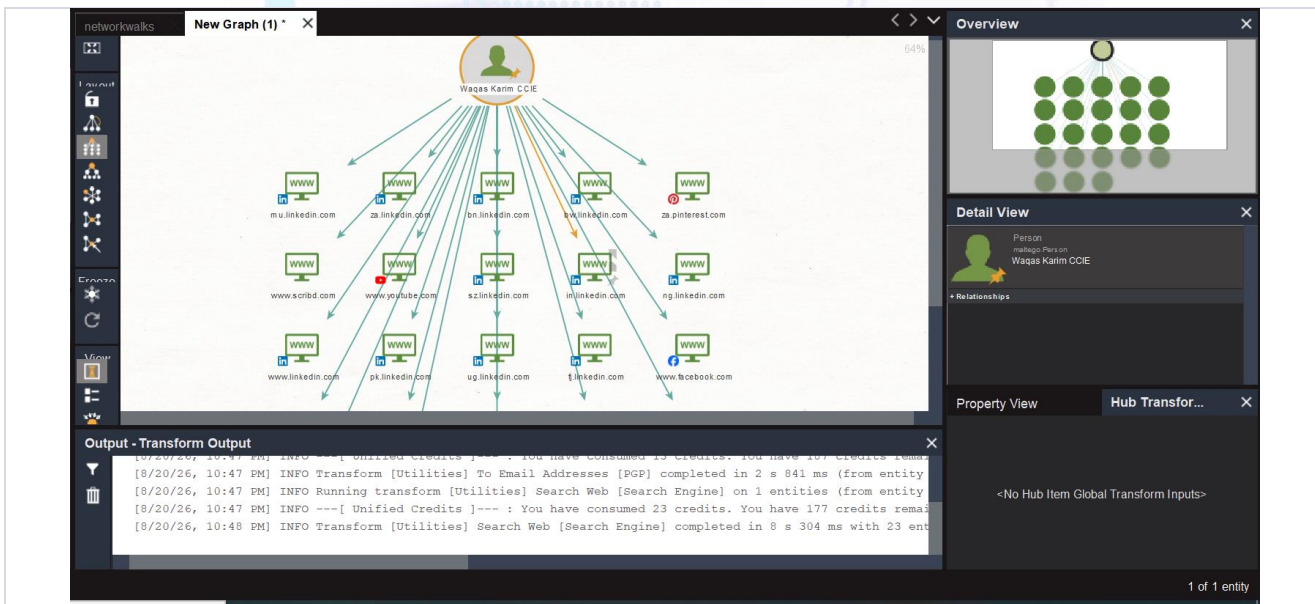


Figure 2.10 — OSINT graph for “Waqas Karim CCIE”, mapping 15 linked web and social-media entities including LinkedIn (multiple regional subdomains), Facebook, Pinterest, YouTube, and Scribd.

The resulting graph fanned out to fifteen linked web properties, dominated by regional LinkedIn subdomains (pk., in., ug., za., bn., sz., ng., fj., mu.linkedin.com) alongside Facebook, Pinterest, YouTube, and Scribd — illustrating how quickly a single named individual's public footprint can be mapped from open sources alone, and why OSINT reconnaissance against employees, not just infrastructure, is a standard part of a penetration-test scoping exercise.

Result

Maltego's transform engine successfully mapped both a domain-level footprint (networkwalks.com → its published contact email) and a person-level footprint (fifteen linked social/web profiles for a named individual), demonstrating how OSINT graphing tools accelerate reconnaissance that would otherwise require many manual searches.

2.3 Email and Host Harvesting with theHarvester (W2-PM4)

theHarvester was used to gather emails, hostnames, and IP addresses for a target domain from public search-engine sources. Running the tool with no arguments confirmed the installed version and required syntax.

```
(userhafs@kali)-[~]
└─$ theHarvester
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* theHarvester
*
*
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-S START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c]
                  [-f FILENAME] [-w WORDLIST] [-a] [-q] [-b SOURCE]
theHarvester: error: the following arguments are required: -d/--domain
```

Figure 2.11 — *theHarvester* 4.10.1 usage/help output, confirming the tool requires a `-d/--domain` argument.

theHarvester was first run against microsoft.com (limit 1000 results, source: Baidu) as a large, well-indexed target to validate the tool was working correctly.

```

└─$ theHarvester -m microsoft.com -l 1000 -b baidu
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****
[*] Target: microsoft.com

[*] Searching Baidu.

[*] No IPs found.

[*] Emails found: 1
viva-noreply@microsoft.com

[*] No people found.

[*] Hosts found: 16
assistants.microsoft.com
c.urs.microsoft.com
fabric.microsoft.com
hxd.research.microsoft.com
mcr.microsoft.com
officecdn.microsoft.com
prod.support.services.microsoft.com
s.windows.microsoft.com
schemas.microsoft.com
securitycopilot.microsoft.com
spf-c.microsoft.com
support.microsoft.com
support.serviceshub.microsoft.com

```

Figure 2.12 — *theHarvester* results for *microsoft.com* via *Baidu*: 1 email address and 16 hosts discovered.

The scan returned one email address (viva-noreply@microsoft.com) and sixteen subdomains/hosts, including assistants.microsoft.com, securitycopilot.microsoft.com, and several regional support and CDN hosts — confirming theHarvester was correctly querying Baidu and parsing results.

The same command was then run against the actual target domain, networkwalks.com.



```
(userh4fsa@kali)~$ theHarvester -d networkwalks.com -l 1000 -b baidu
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: networkwalks.com

[*] Searching Baidu.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] No hosts found.

(userh4fsa@kali)~$
```

Figure 2.13 — theHarvester results for networkwalks.com via Baidu: no IPs, emails, people, or hosts found.

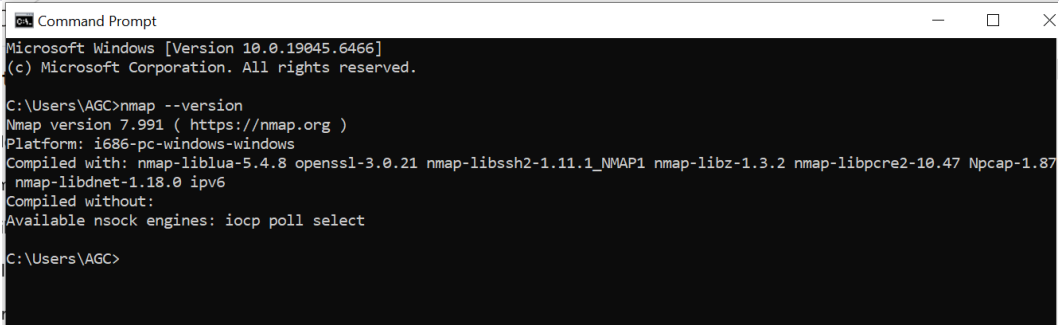
This run returned no results at all from the Baidu source. This is an expected limitation rather than a tool failure: theHarvester's results depend entirely on how well a target is indexed by the specific source queried, and a smaller regional training-academy site is far less likely to appear in Baidu's index than a global site like microsoft.com. This is a useful, realistic finding in its own right — in a real engagement it would prompt trying additional sources (e.g. Google, Bing, crt.sh, or DNS-based sources) rather than concluding the domain has no discoverable footprint.

Result

theHarvester was validated against a high-profile domain (microsoft.com), successfully returning email and host data, before being run against the target domain networkwalks.com, which returned no results from the Baidu source — highlighting that harvester results are source- and target-dependent, and that multiple sources should be queried during real reconnaissance.

2.4 Network Scanning with Zenmap (W2-PM5)

Zenmap, the official GUI front-end for Nmap, was installed on the Windows host machine to perform network scanning and host discovery. Installation was verified from the command line before opening the GUI.



```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\AGC>nmap --version
Nmap version 7.991 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-1.3.0.21 nmap-libssh2-1.11.1_NMAP1 nmap-libz-1.3.2 nmap-libpcap-1.87
nmap-libnet-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\AGC>
```

Figure 2.14 — Verifying the Nmap 7.991 installation from Windows Command Prompt with nmap --version.

With Nmap confirmed installed, Zenmap was opened and a Quick Scan profile was run against a target on the local network (target masked for privacy).

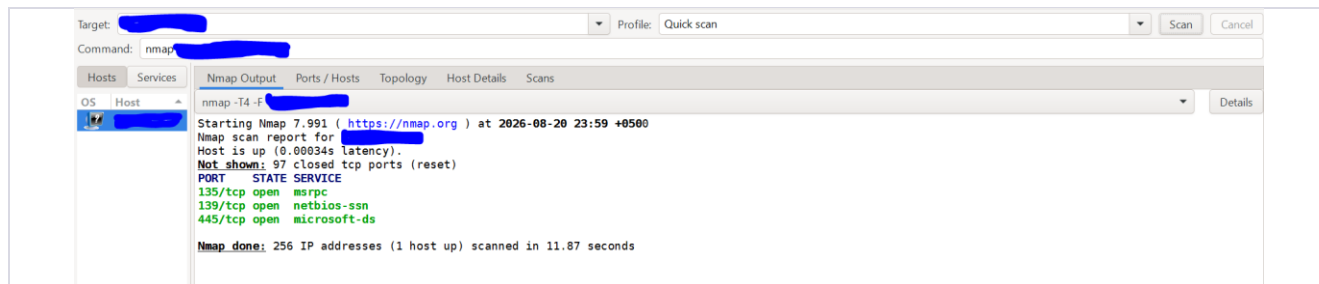


Figure 2.15 — Zenmap Quick Scan (`nmap -T4 -F`) results: host up, with ports 135, 139, and 445 open.

The Quick Scan (`nmap -T4 -F`, a fast scan of the most common ports) found the host up and reachable, with three TCP ports open: 135 (MSRPC), 139 (NetBIOS Session Service), and 445 (Microsoft-DS/SMB) — identifying the target as a Windows host with file/print-sharing services exposed on the network. A Ping Scan (`nmap -sn`, host discovery only, no port scan) was then run to confirm the host was reachable without probing any ports.

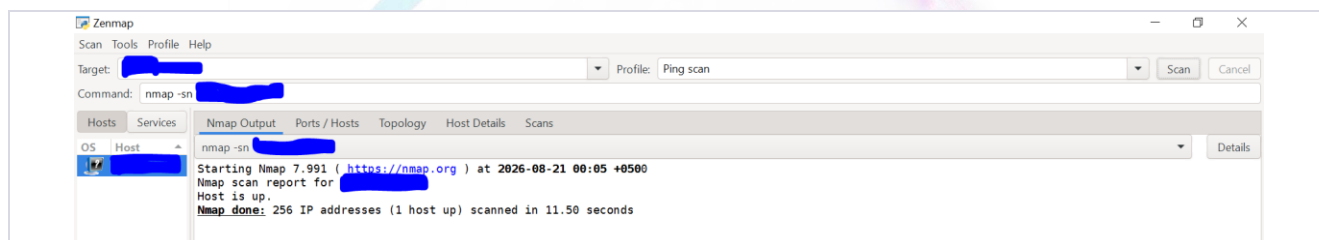


Figure 2.16 — Zenmap Ping Scan (`nmap -sn`) confirming the target host is up, with no ports probed.

Both scans completed in under twelve seconds and swept the full local /24 address range (256 IP addresses), confirming exactly one host responding at the scanned address. From a security perspective, an exposed SMB service (port 445) alongside NetBIOS (139) and RPC (135) is a classic indicator that a Windows machine could be vulnerable to SMB-based attacks (e.g. relay, enumeration, or unpatched exploits) if not properly firewalled from untrusted networks — exactly the kind of finding this reconnaissance phase is designed to surface.

Result

Zenmap's Quick Scan identified an active Windows host on the local network with SMB/NetBIOS/RPC ports (445, 139, 135) open, and a follow-up Ping Scan confirmed host availability independent of port state — completing the network-scanning portion of this week's reconnaissance work.

3. Challenges Faced

- Inconsistent theHarvester results: the most instructive challenge of this project was theHarvester returning zero results for the actual target domain (networkwalks.com) while the same command against microsoft.com returned rich data. This required recognizing that an empty result reflects a source/target indexing limitation rather than a tool or configuration error, as documented in §2.3.
- Interpreting layered tool output: each Kali footprinting tool in §2.1 surfaces a different, narrow slice of information (registration, web stack, DNS, WAF), so building a complete picture of the target required deliberately cross-referencing results across all six tools rather than treating any single tool's output as sufficient on its own.
- Managing Maltego transform credits: Maltego's free Community Edition uses a consumable transform-credit system (visible in the Output log in Figures 2.9 and 2.10), which meant transforms had to be chosen deliberately rather than run indiscriminately.

4. Lessons Learned

Carrying out this week's reconnaissance and footprinting work reinforced several practical lessons:

- Footprinting is most effective as a layered process: no single tool gives a complete picture, but combining WHOIS, web-stack fingerprinting, DNS enumeration, WAF detection, OSINT graphing, and port scanning together builds a genuinely useful target profile.
- A web application firewall (as detected by wafw00f in §2.1.5) is itself an important reconnaissance finding, since it directly affects how later, more active testing phases should be paced and structured.
- OSINT tools like Maltego and theHarvester are only as good as the sources they query — a “no results” finding is meaningful information about a target's public footprint, not a dead end, and should prompt trying additional sources.
- Passive/light scanning techniques (WHOIS, DNS, ping scans) should generally be exhausted before moving to more active scanning (full port scans, service/version detection), reflecting the standard reconnaissance-before-scanning-before-exploitation flow of a penetration test.
- Open SMB-related ports (135, 139, 445) discovered via Zenmap are a well-known high-value finding, since they are frequently the entry point for lateral movement and known Windows exploits if left exposed on an untrusted network segment.

5. Conclusion

All items in scope for this Week 02 project were completed: the target domain networkwalks.com was footprinted using six Kali Linux command-line tools, OSINT relationships were mapped for both the domain and a named individual using Maltego, email and host data was harvested using theHarvester (validated against a high-traffic domain and then run against the target), and a local network target was scanned for open ports and host availability using Zenmap.

Together, these four modules demonstrate the full breadth of passive and light-active reconnaissance available before any exploitation is attempted: registration and DNS data, web technology and WAF fingerprinting,

OSINT-derived email and social footprints, and network-level port and host discovery. This mirrors the reconnaissance phase of a real penetration-testing engagement, and the findings from this week — particularly the ModSecurity WAF and the exposed SMB ports — would directly inform the planning of any later, more active testing phase of this program.

6. References

- Kali Linux — Official Website and Tool Documentation, <https://www.kali.org>
- Nmap — Official Documentation, <https://nmap.org/book/man.html>
- Zenmap — Official Documentation, <https://nmap.org/book/zenmap.html>
- Maltego — Official Website and Documentation, <https://www.maltego.com>
- theHarvester — GitHub Repository, <https://github.com/laramies/theHarvester>
- WAFW00F — GitHub Repository, <https://github.com/EnableSecurity/wafw00f>
- dnsrecon — GitHub Repository, <https://github.com/darkoperator/dnsrecon>
- Networkwalks Academy, <https://networkwalks.com>
- Networkwalks Academy — Cybersecurity Professional Program, Week 02 Project Brief (internal)